

The Association between Gene Expression Profiling and Regression Rate of Posterior Uveal Melanoma following Proton Beam Irradiation

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Keywords

Uveal melanoma · Gene expression profile · Preferentially expressed antigen in melanoma · Regression rate · Proton beam irradiation

Abstract

Introduction: This study evaluated the association between gene expression profiling (GEP), PRAME (preferentially expressed antigen in melanoma), and regression rate of uveal melanoma after proton beam irradiation (PBI). **Methods:** A retrospective review of uveal melanoma patients treated with PBI between 2013 and 2021, with GEP results and at least 3 post-radiation ultrasound measurements, was conducted. Patients with local recurrences were excluded. Regression rates were analyzed using a linear mixed model to predict percentage change in thickness from baseline. Cox regression was conducted to determine whether slow or fast regression, based on the median regression rate at 18 months, correlates with metastasis risk. **Results:** The study included 106 patients, with GEP classifications of 1A in 43.4%, 1B in 25.5%, and 2 in 31.1%. Overall, the mean change in tumor thickness was 20.9%, 35.1%, 51.4%, and 59.3% at 1 year, 2 years, 4 years, and 6 years, respectively. No differences in regression rates between GEP classes and PRAME expression were found through 72 months post-PBI. The

median regression at 18 months was 27.1%. Slow and fast regression was not associated with the risk of metastasis. **Conclusion:** No association between GEP, PRAME, and regression rate was found through 72 months post-PBI.

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Introduction

After the Collaborative Ocular Melanoma Study (COMS) group showed no long-term survival benefit to enucleation over brachytherapy, radiation therapy became the standard treatment for most cases of uveal melanoma [1, 2]. The two most common forms of irradiation, brachytherapy, and proton beam radiotherapy have excellent local tumor control rates of more than 90% [3–5]. Despite successful treatment, most tumors do not completely disappear even 54 months post-irradiation [6]. Indeed, rapid regression after irradiation may be associated with poor prognosis [7–12] in uveal melanoma patients.

Gene expression profiling (GEP) using a commercial assay (DecisionDx-UM, Castle Biosciences) to analyze 12 discriminating genes and 3 control genes from tumor samples is a validated predictor of metastasis [13, 14]. Patients with class 2 tumors have a more than 70% chance

of developing metastasis within 5 years of diagnosis [15]. Higher cell proliferation rates, epithelioid cell type, and monosomy 3, known factors associated with poor survival, have been reported to be associated with class 2 tumors [13], as well as faster-regressing tumors [12, 16]. However, several studies have found no association between GEP and regression rates [17–19] or found faster regression rates in class 1 tumors compared to class 2 tumors in short-term follow-up after treatment [20, 21].

Most published studies evaluated tumor regression rates following brachytherapy with a relatively short follow-up period of 3–24 months [7, 10–12, 18–24]. Still, many tumors continue to regress after 24 months [6, 8, 25]. Therefore, this study aimed to evaluate the impact of GEP and PRAME (preferentially expressed antigen in melanoma), another biomarker of metastatic risk [26, 27], on long-term tumor regression rates after treatment with proton beam irradiation (PBI) in patients with choroidal melanoma.

Methods

This study was approved by the Mass General Brigham Institutional Review Board and conducted under Protocol 2019P000606, Epidemiologic Studies of Uveal Melanoma, and Protocol 2019P000285, the Uveal Melanoma Repository. A retrospective review was performed to identify choroidal or ciliochoroidal melanoma patients who were enrolled in the Uveal Melanoma Repository (UMR) and treated with PBI between 2013 and 2021. Patients were included in the analysis if GEP test results (DecisionDx-UM, Castle Biosciences, Friendswood, TX) and at least 3 tumor measurements post-irradiation by either B-scan or ultrasound biomicroscopy were available. Seven patients with recurrent uveal melanoma at baseline or who developed local recurrence during follow-up visits were excluded. PRAME results were only available beginning in 2016.

Demographic data, tumor characteristics at the baseline examination (tumor shape, ciliary body involvement, extrascleral extension, largest basal diameter [LBD], thickness), and total radiation dose were obtained from medical records. Baseline maximal tumor thickness was assessed by both standardized A-scan and contact B-scan. Ultrasound biomicroscopy was used for anteriorly located tumors. All measurements were performed by experienced ultrasonographers following standard clinical protocols at the Massachusetts Eye and Ear. Tumor measurements were obtained from medical records at every visit. The radiation dose was selected based

on pretreatment tumor size and location as described previously [28]. In brief, tumors ≤ 15 mm in LBD, ≤ 5 mm in thickness, located within 1 disc diameter of the optic nerve and/or fovea were treated with 50 Gy (RBE, relative biological effectiveness) given in 5 fractions. Otherwise, the standard dose of 70 Gy (RBE) was used. Clinical outcomes including local recurrence, metastasis, and vital status at the end of the study were noted. Survival status was ascertained from available medical records, the Social Security Death Index (<http://www.ancestry.com>), the National Death Index (NDI), or contact with the patient's next of kin.

Statistical Analysis

The primary endpoint evaluated was the percentage change in tumor thickness from the pre-treatment measure to measures over time and at designated time-points after PBI. Because these repeated measures data were unbalanced with variable time intervals, direct comparison of changes in tumor thickness was not feasible, and thus a linear mixed model was used. To predict the percentage change in tumor thickness over time and the functional form of the curve, various parameterizations of time and interactions between GEP class, PRAME, and response over time were explored. The optimum model was selected as the one that minimized the Akaike information criterion.

Secondary endpoints included the evaluation of factors influencing changes in tumor thickness and assessing the association between rate of regression and metastasis. Cox regression analysis was used to assess whether fast or slow regression rate is associated with metastasis.

Differences in patient and tumor characteristics by tumor class were evaluated using the Fisher's exact test for categorical variables and Mann-Whitney U test for continuous variables. $p < 0.05$ was considered statistically significant. Statistical analysis was performed using Stata (software version 18; StataCorp LLC, College Station, TX).

Results

A total of 106 patients were included in the study. A summary of patient and tumor characteristics is shown in Table 1. Overall, 41 patients (38.7%) were female and the median age at diagnosis was 60.5 years. The tumors were classified as class 1A in 46 patients (43.4%), class 1B in 27 patients (25.5%), and class 2 in 33 patients (31.1%). Class 1B patients had a significantly longer follow-up period than class 2 patients ($p = 0.01$). There were no differences

Table 1. Patients and tumor characteristics

Characteristics	Class 1A (n = 46)	Class 1B (n = 27)	Class 2 (n = 33)	p value
Age, median (range), years	60.5 (36.3–85.3)	55.2 (19.2–77.6)	66.2 (38.9–81.9)	0.07
Female, n (%)	12 (26.1%)	12 (44.4%)	17 (51.5%)	0.05
Right eye, n (%)	20 (43.5%)	14 (53.9%)	16 (48.5%)	0.69
Follow-up, median (range), months	49.1 (19.4–112.5)	56.8 (27.8–100.2)	42.1 (10.5–106.4)	0.03*
Total number of US, median (range)	9 (4–19)	10 (4–14)	8 (4–13)	0.09
Interval between US, median (range), months	6.5 (3.7–19.9)	6.3 (4.7–10.2)	6.1 (2.8–10.0)	0.03*
PRAME, n (%)				
Negative	28 (60.9%)	12 (44.5%)	14 (42.4%)	0.22
Positive	3 (6.5%)	6 (22.2%)	7 (21.2%)	
N/A	15 (32.6%)	9 (33.3%)	12 (36.4%)	
Tumor shape, n (%)				
Dome	32 (69.6%)	21 (77.8%)	24 (72.7%)	0.03*
Mushroom	11 (23.9%)	3 (11.1%)	4 (12.1%)	
Diffuse	3 (6.5%)	0 (0%)	0 (0%)	
Irregular	0 (0%)	3 (11.1%)	5 (15.2%)	
Ciliary body involvement, n (%)	6 (13.0%)	3 (11.1%)	10 (30.3%)	0.08
Extrasccleral extension, n (%)	1 (2.2%)	0	4 (12.1%)	0.07
LBD, median (range), mm	10.1 (6.5–16.5)	10.4 (6.7–17.1)	12.4 (7.5–19.8)	0.04*
Baseline thickness, median (range), mm	3.5 (1.7–11.0)	3.7 (1.5–9.7)	5.4 (1.9–12.5)	0.21
Radiation dose, n (%)				
50 Gy	20 (43.5%)	9 (33.3%)	5 (15.2%)	0.02*
70 Gy	26 (56.5%)	18 (66.7%)	28 (84.9%)	

US, ultrasound; PRAME, preferentially expressed antigen in melanoma; LBD, largest basal diameter; Gy: RBE, relative biological effectiveness. *Statistically significant *p* value <0.05.

in follow-up time between patients with class 1A versus class 1B tumors ($p = 0.27$) or class 1A versus class 2 tumors ($p = 0.06$). The number of total tumor measurements was not significantly different between each class. However, the interval between each measurement was shorter in class 2 (median time of 6.1 months) compared with both class 1 categories (6.5 and 6.3 months in class 1A and class 1B, respectively; $p = 0.03$). Class 2 tumors were significantly larger in baseline basal diameter than class 1 tumors (median of 12.4 mm, 10.4 mm, and 10.1 mm in class 2, class 1B, and class 1A, respectively; $p = 0.04$) but did not differ in baseline thickness ($p = 0.21$). More patients with class 2 tumors (84.9%) received a treatment dose of 70 Gy (RBE) compared to those with class 1A tumors (56.5%) and class 1B tumors (66.7%).

The optimum model determined to predict the percentage reduction in tumor thickness from baseline by GEP class included a cubic and quadratic term for time, baseline tumor thickness, a random intercept, and a

random coefficient. The beta coefficient for time after radiation was $\beta = -1.61$ (95% CI: $-1.86, -1.37$; $p < 0.001$) and for baseline tumor thickness was $\beta = -1.14$ (95% CI: $-1.91, -0.37$; $p = 0.004$). We included total dose in initial mixed models, but it was not significantly associated with regression rates ($\beta = 3.22$, 95% CI: $-2.25, 8.68$; $p = 0.25$). Additionally, the Akaike information criterion indicated that the model had a better fit without inclusion of dose. As a result, radiation dose was not included in the final model.

The percentage changes in each GEP class are shown in Figure 1 and online supplementary Figure 1 (for all online suppl. material, see <https://doi.org/10.1159/000542397>). Regression values at 6, 12, 24, 48, and 72 months for class 1A tumors were 10.2%, 18.2%, 31.2%, 47.5%, and 54.3%, respectively; class 1B tumors regressed 14.9%, 25.5%, 40.8%, 56.5%, and 67.4%; and class 2 tumors regressed 11.0%, 20.6%, 35.6%, 52.6%, and 59.2%. There were no significant differences in regression rates between GEP classes at any point in time through

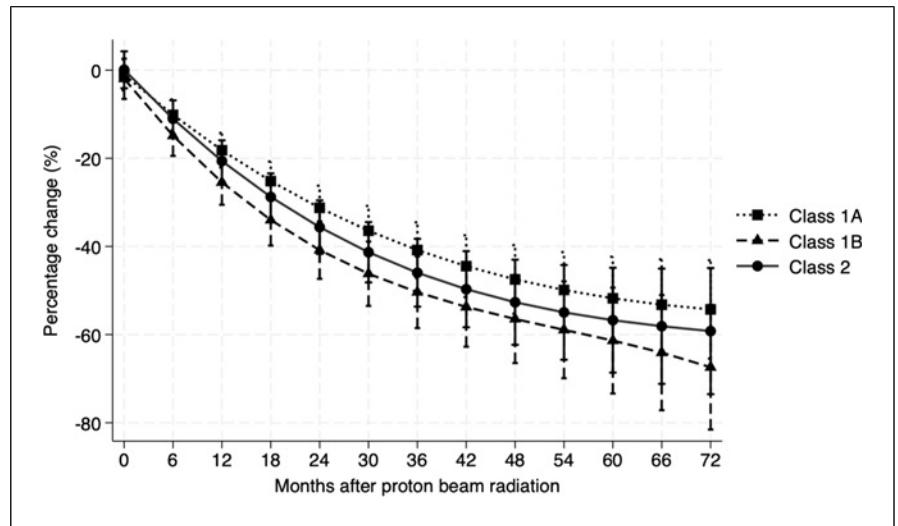


Fig. 1. Predicted percentage change in tumor thickness from baseline thickness, by GEP class.

72 months post-PBI ($\beta = -0.92$, 95% CI: -6.68 , 4.85 for class 1B tumors compared to class 1A tumors and $\beta = 1.13$, 95% CI: -4.37 , 6.62 for class 2 tumors compared to class 1A tumors). Subgroup analysis was performed by categorizing patients according to their baseline tumor thickness. Three groups were defined to provide consistency with the study by Demirci et al. [8]: thickness ≤ 3 mm, thickness >3 but ≤ 8 mm, and thickness >8 mm. No differences in regression rates between GEP classes were observed (shown in Fig. 2 and online suppl. 2).

Among 70 patients who had PRAME results, no significant differences in tumor regression rates were found between patients with class 1 PRAME negative, class 1 PRAME positive, and class 2 tumors at any point in time through 72 months (shown in Fig. 3 and online suppl. 3). Subgroup analysis to further evaluate PRAME by baseline tumor thickness categories was not performed due to the small sample size.

At the end of the study, metastasis was diagnosed in 3 patients (6.5%) with class 1A tumors, 1 (3.7%) with class 1B tumors, and 17 (51.5%) with class 2 tumors ($p < 0.01$). Death from any cause occurred in 2 (4.4%) patients with class 1A tumors, 2 (7.4%) with class 1B tumors, and 10 (30.3%) with class 2 tumors ($p < 0.01$). To evaluate the association between regression rate and metastasis, patients were divided into 2 groups based on comparing the individual percentage change in tumor thickness to the overall median percentage change at 18 months. Tumors that had a lower percentage change than the median value were considered slow regressors; otherwise, they were considered fast regressors. The 24-month percentage change ($n = 19$) or the 12-month percentage change ($n = 7$) was used when an 18-month measure was unavailable.

Additionally, for 4 patients for whom metastasis had occurred before 18 months after PBI, the 12-month percentage change was used. The cutoff at 18 months was chosen because more than half of the change in tumor thickness occurred within this timeframe. Additionally, the peak regression in tumor thickness had occurred in half of the cohort. The characteristics of slow and fast regressors are presented in Table 2. Fast regressors had shorter follow-up times and a higher prevalence of tumors with mushroom shape compared to slow regressors, but these differences were not statistically significant. Ten (17.5%) patients in the slow regression group and 11 (22.9%) patients in the fast regression group developed metastasis ($p = 0.62$). The risk of metastasis was not increased in fast regressors compared to slow regressors (crude HR [95% CI]: 1.38 [0.54 – 3.01]). Significant risk factors in a multivariable model were class 2 tumors and baseline LBD (shown in online suppl. 4).

Discussion

As metastasis is the leading cause of death in uveal melanoma patients [29], attempts to find precise predictors of disease progression has long been sought. Patient characteristics, tumor features, histopathology and genetic biomarkers are essential tools to accurately determine prognosis. GEP has been validated as a highly accurate predictor of metastatic risk [13]. Tumor response to radiation therapy has also been implicated as an indicator of metastatic risk [7–10]. Previous studies have suggested that rapidly regressing tumors are more likely to metastasize than those that regress more slowly and are

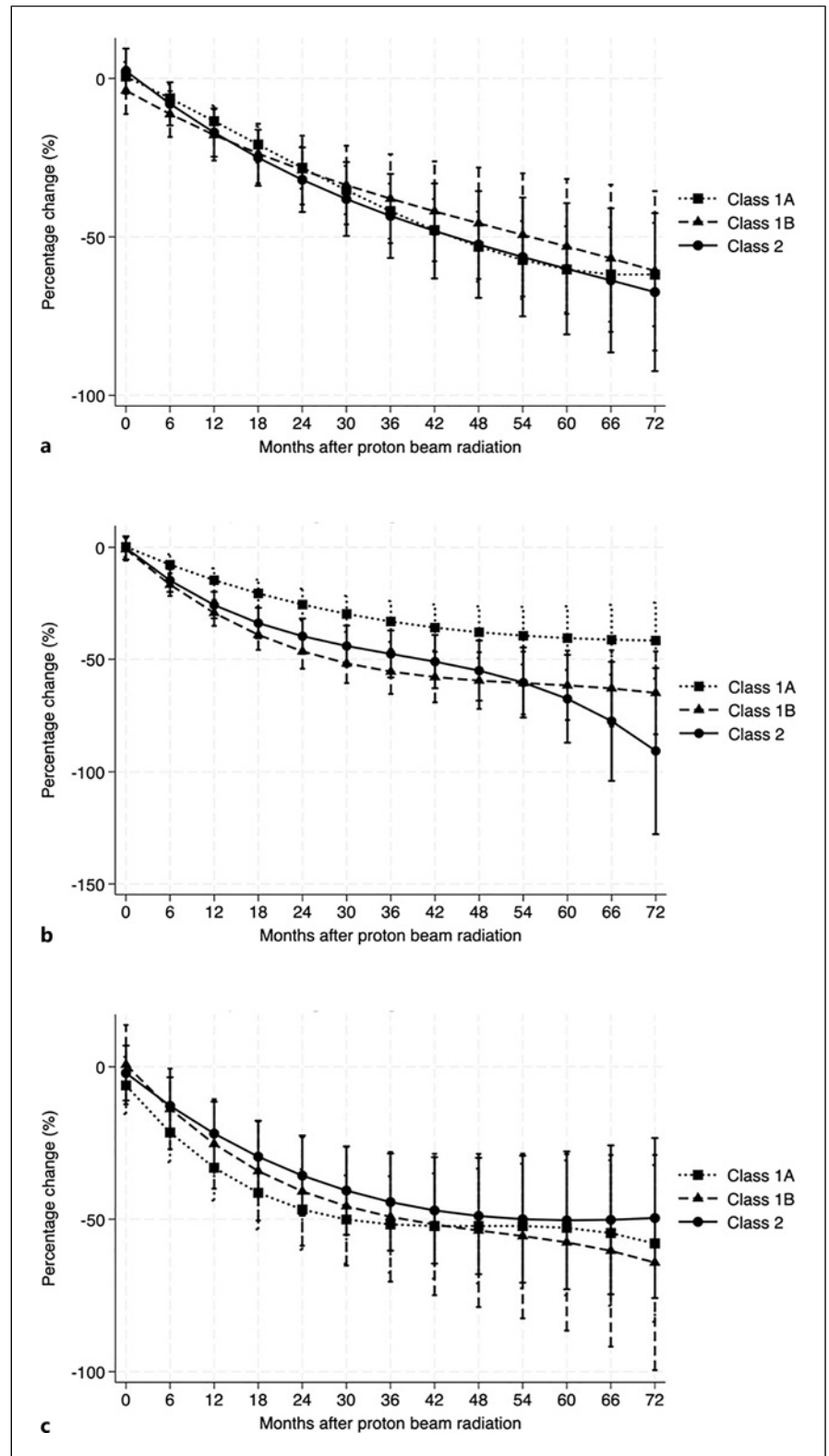


Fig. 2. Predicted percentage change in tumor thickness from baseline thickness, by baseline thickness and GEP class. **a** Baseline tumor thickness ≤ 3 mm. **b** Baseline tumor thickness > 3 mm and ≤ 8 mm. **c** Baseline tumor thickness > 8 mm.

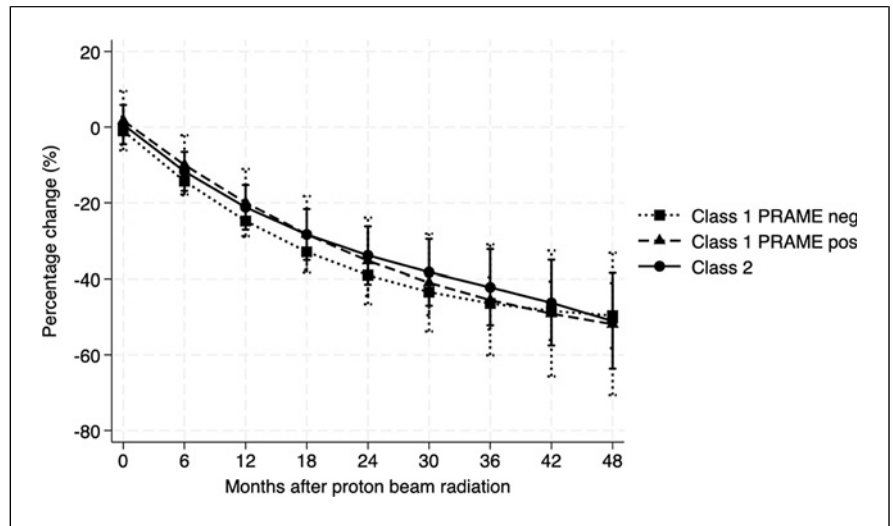


Fig. 3. Predicted percentage change in tumor thickness from baseline thickness, by GEP class and PRAME expression.

Table 2. Patients and tumor characteristics of slow and fast regressors

Characteristics	Slow regressor (n = 57)	Fast regressor (n = 48)	p value
Age, median (range), years	63.0 (37.3–77.8)	59.3 (19.2–85.3)	0.22
Female, n (%)	22 (38.6%)	18 (37.5%)	1.0
Right eye, n (%)	29 (50.9%)	21 (43.8%)	0.56
Follow-up, median (range), months	50.7 (10.5–112.5)	43.5 (16.6–106.4)	0.38
Total number of US, median (range)	9 (4–19)	9 (4–14)	0.45
Interval between US, median (range), months	6.2 (2.8–19.9)	6.2 (3.7–10.0)	0.91
GEP, n (%)			
1A	28 (49.1%)	17 (35.4%)	0.045
1B	9 (15.8%)	18 (37.5%)	
2	20 (35.1%)	13 (27.1%)	
PRAME, n (%)			
Negative	29 (50.9%)	24 (50.0%)	0.64
Positive	7 (12.3%)	9 (18.8%)	
N/A	21 (36.8%)	15 (31.3%)	
Tumor shape, n (%)			
Dome	44 (77.2%)	32 (66.7%)	0.25
Mushroom	6 (10.5%)	12 (25.0%)	
Diffuse	2 (3.5%)	1 (2.1%)	
Irregular	5 (8.8%)	3 (6.3%)	
Ciliary body involvement, n (%)	11 (19.3%)	8 (16.7%)	0.80
Extrasccleral extension, n (%)	4 (7.0%)	1 (2.1%)	0.37
LBD, median (range), mm	10.9 (6.5–19.8)	11.2 (6.7–18.0)	0.92
Baseline thickness, median (range), mm	3.6 (1.5–12.5)	4.6 (1.9–10.1)	0.17
Radiation dose, n (%)			
50 Gy	21 (36.8%)	12 (25.0%)	0.21
70 Gy	36 (63.2%)	36 (75.0%)	

US, ultrasound; GEP, genetic expression profiling; PRAME, preferentially expressed antigen in melanoma; LBD, largest basal diameter; Gy: RBE, relative biological effectiveness.

associated with monosomy 3 status [7–12]. Therefore, it is reasonable to investigate a possible relationship between GEP classification and tumor regression after radiotherapy.

The Collaborative Ocular Oncology Group (COOG) found that class 2 tumors were associated with greater tumor thickness, ciliary body involvement, mixed or epithelioid cell type, and monosomy 3 status, which are all known clinical risk factors for metastasis [30]. Tumors with these features are likely to have higher cell cycling rates and tend to grow faster [31, 32], which in theory should have made these tumors more radiosensitive [33, 34]. In our study, which had longer follow-up data available than previous studies, tumors regressed by 20.9%, 35.1%, 51.4%, and 59.3% of baseline thickness (percent change from baseline thickness) at 1 year, 2 years, 4 years, and 6 years, respectively. We did not find an association between GEP and tumor regression rate after PBI, consistent with the results of investigations of tumor regression after brachytherapy [17–19]. Two studies found no significant difference in regression rate at 6 months after I-125 brachytherapy [18, 19], while two other studies reported faster regression rates of class 1 tumors but only at 3 or 6 months after treatment and not at later timepoints [20, 21]. Investigations of an association between regression rate and GEP after PBI are lacking. A study from Chappell et al. is the only published report evaluating this relationship after PBI. They evaluated the change in tumor thickness at 24 months in 197 patients and found no difference in regression rates between class 1 and class 2 tumors [17].

Most of these studies did not subdivide class 1 tumors into class 1A and 1B groups and none evaluated the association of PRAME, which has been shown to improve metastatic risk prediction in class 1 tumors [27, 35], and tumor regression rates. In this study, we completed an analysis of PRAME in a subgroup of patients who had results from PRAME testing and found no significant differences in regression rates between each class according to GEP and PRAME status. To the best of the authors' knowledge, this is one of the only studies to have analyzed both 1A versus 1B and 1PRAME+ and 1PRAME–.

When subgrouping the patients based on baseline tumor thickness, no difference in regression rate was observed between GEP classes. However, although not statistically significant, the most noticeable difference in regression rates was observed in medium-size tumors with baseline thickness between 3 and 8 mm. The less noticeable difference in the small size group might be due to the floor effect while the large tumors might contain

more proliferating cells and be more susceptible to radiation regardless of tumor class.

Regression rates after radiation therapy also depend on other factors including the ability of the tumor to remove dead cells, dynamic balance between different tumor components, and the reaction of host against residual cells [8, 25]. We cannot exclude the possibility that the influence of baseline thickness in the regression model may have some effect from tumor-associated hemorrhage. Given that larger tumors have a higher incidence of intratumoral hemorrhage, this could potentially impact regression rates. However, it is unlikely that rates of hemorrhage would be affected by GEP class. It is also possible that total radiation dose may also influence regression rates. However, we did not find inclusion of dose improved the predictive value of the model for regression rates. However, the sample size was limited, and the correlation between tumor size and dose selection complicates the assessment of the independent effect of dose on regression rates.

Our results confirm the validity of GEP as a strong predictor of metastasis, as class 2 tumors were significantly associated with the risk of metastasis. However, we found no evidence of a significant relationship between regression rates and GEP class or of a correlation between regression rates and risk of metastasis. The limitations of this study include the relatively small sample size, especially the number of patients with PRAME results. Given the retrospective nature of this study, intra- and inter-observer reliability of tumor measurement could not be evaluated. However, all measurements were done by a limited number of experienced ultrasonographers. Finally, only one tumor dimension was used in this study. Rashid et al. [36] reported different regression rates in tumors with differently shaped tumors and suggested that cross-sectional area should be considered when evaluating prognostic value of regression rates.

Statement of Ethics

Mass General Brigham Institutional Review Board approved the study, conducted under Protocol 2019P000606, Epidemiologic Studies of Uveal Melanoma, and Protocol 2019P000285, the Uveal Melanoma Repository. Informed consent was exempted by the Mass General Brigham Institutional Review Board (IRB) for this study.

Conflict of Interest Statement

Disorn Suwajanakorn, Anne Marie Lane, and Frances Wu: none. Evangelos S. Gragoudas: consultant: Aura Biosciences. Ivana K. Kim: funding: Allergan and consultant: Genentech, Biophytis, and Kodiak Sciences.

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Author Contributions

Disorn Suwajanakorn: conceptualization, formal analysis, investigation, methodology, visualization, and writing – original draft. Anne Marie Lane: formal analysis, investigation, methodology, validation, and writing – review and editing. Frances Wu:

conceptualization and writing – review and editing. Evangelos S. Gragoudas: resources and writing – review and editing. Ivana K. Kim: conceptualization, methodology, project administration, supervision, validation, and writing – review and editing. All authors are aware of and agree to the content of this paper and agree to be listed as authors.

Data Availability Statement

The data that support the findings of this study are not publicly available due to the inclusion of information that could compromise the privacy of research participants. However, they are available from the corresponding author, Dr. Ivana K. Kim, upon reasonable request.

References

- 1 Kaliki S, Shields CL. Uveal melanoma: relatively rare but deadly cancer. *Eye*. 2017;31(2): 241–57. <https://doi.org/10.1038/eye.2016.275>
- 2 Salowe R, O'Brien JM. The collaborative ocular melanoma study randomized trial of iodine 125 brachytherapy for choroidal melanoma. *Ophthalmology*. 2020;127(4s): S146–s7.
- 3 Gragoudas E, Li W, Goitein M, Lane AM, Munzenrider JE, Egan KM. Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. *Arch Ophthalmol*. 2002;120(12):1665–71. <https://doi.org/10.1001/archophth.120.12.1665>
- 4 Jampol LM, Moy CS, Murray TG, Reynolds SM, Albert DM, Schachat AP, et al. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 Years after brachytherapy. COMS report No. 19. *Ophthalmology*. 2020;127(4s): S148–57. <https://doi.org/10.1016/j.ophtha.2020.01.032>
- 5 Buonanno F, Conson M, de Almeida Ribeiro C, Oliviero C, Itta F, Liuzzi R, et al. Local tumor control and treatment related toxicity after plaque brachytherapy for uveal melanoma: a systematic review and a data pooled analysis. *Radiother Oncol*. 2022;166:15–25. <https://doi.org/10.1016/j.radonc.2021.11.007>
- 6 Cruess AF, Augsburger JJ, Shields JA, Brady LW, Markoe AM, Day JL. Regression of posterior uveal melanomas following cobalt-60 plaque radiotherapy. *Ophthalmology*. 1984;91(12):1716–9. [https://doi.org/10.1016/s0161-6420\(84\)34087-8](https://doi.org/10.1016/s0161-6420(84)34087-8)
- 7 Augsburger JJ, Gamel JW, Shields JA, Markoe AM, Brady LW. Post-irradiation regression of choroidal melanomas as a risk factor for death from metastatic disease. *Ophthalmology*. 1987;94(9):1173–7. [https://doi.org/10.1016/s0161-6420\(87\)33310-x](https://doi.org/10.1016/s0161-6420(87)33310-x)
- 8 Demirci H, Saponara F, Khan A, Niziol LM, Lee C, Hayman JA, et al. Regression rate of posterior uveal melanomas following iodine-125 plaque radiotherapy. *Middle East Afr J Ophthalmol*. 2015;22(1):103–7. <https://doi.org/10.4103/0974-9233.148358>
- 9 Glynn RJ, Seddon JM, Gragoudas ES, Egan KM, Hart LJ. Evaluation of tumor regression and other prognostic factors for early and late metastasis after proton irradiation of uveal melanoma. *Ophthalmology*. 1989;96(10):1566–73. [https://doi.org/10.1016/s0161-6420\(89\)32685-6](https://doi.org/10.1016/s0161-6420(89)32685-6)
- 10 Kaiserman I, Anteby I, Chowers I, Blumen-thal EZ, Kliars I, Pe'er J. Post-brachytherapy initial tumour regression rate correlates with metastatic spread in posterior uveal melanoma. *Br J Ophthalmol*. 2004;88(7):892–5. <https://doi.org/10.1136/bjo.2003.036285>
- 11 Marathe OS, Wu J, Lee SP, Yu F, Burgess BL, Leu M, et al. Ocular response of choroidal melanoma with monosomy 3 versus disomy 3 after iodine-125 brachytherapy. *Int J Radiat Oncol Biol Phys*. 2011;81(4):1046–8. <https://doi.org/10.1016/j.ijrobp.2010.07.016>
- 12 Shields CL, Bianciotto C, Rudich D, Materin MA, Ganguly A, Shields JA. Regression of uveal melanoma after plaque radiotherapy and thermotherapy based on chromosome 3 status. *Retina*. 2008;28(9):1289–95. <https://doi.org/10.1097/IAE.0b013e31817f7b3e>
- 13 Dogrusöz M, Jager MJ. Genetic prognostication in uveal melanoma. *Acta Ophthalmol*. 2018;96(4):331–47. <https://doi.org/10.1111/aos.13580>
- 14 Onken MD, Worley LA, Tuscan MD, Harbour JW. An accurate, clinically feasible multi-gene expression assay for predicting metastasis in uveal melanoma. *J Mol Diagn*. 2010;12(4):461–8. <https://doi.org/10.2353/jmol dx.2010.090220>
- 15 Aaberg TM, Covington KR, Tsai T, Shildkrot Y, Plasseraud KM, Alsina KM, et al. Gene expression profiling in uveal melanoma: five-year prospective outcomes and meta-analysis. *Ocul Oncol Pathol*. 2020;6(5): 360–7. <https://doi.org/10.1159/000508382>
- 16 Lee CS, Lee SC, Lee K, Kwon HJ, Yi JH, Cho A. Regression of uveal melanoma after ru-106 brachytherapy and thermotherapy based on metabolic activity measured by positron emission tomography/computed tomography. *Retina*. 2014;34(1):182–7. <https://doi.org/10.1097/IAE.0b013e31829479fb>
- 17 Chappell MC, Char DH, Cole TB, Harbour JW, Mishra K, Weinberg VK, et al. Uveal melanoma: molecular pattern, clinical features, and radiation response. *Am J Ophthalmol*. 2012;154(2):227–32.e2. <https://doi.org/10.1016/j.ajo.2012.02.022>
- 18 Corrêa ZM, Augsburger JJ. Relationship between rate of posterior uveal melanoma flattening following plaque radiotherapy and gene expression profile class of tumor cells. *Invest Ophthalmol Vis Sci*. 2014;55(1):556–9. <https://doi.org/10.1167/iov.13-13381>
- 19 Gupta K, McCannel CA, Kamrava M, Lamb J, Almazan RD, McCannel TA. Tumor-height regression rate after brachytherapy between choroidal melanoma gene expression profile classes: effect of controlling for tumor height. *Graefes Arch Clin Exp Ophthalmol*. 2016;254(7):1371–8. <https://doi.org/10.1007/s00417-016-3305-2>
- 20 Rao RC, Khan M, Badiyan SN, Harbour JW. Gene expression profiling and regression rate of irradiated uveal melanomas. *Ophthalmic Surg Lasers Imaging Retina*. 2015;46(3):333–7. <https://doi.org/10.3928/23258160-20150323-06>
- 21 Mruthyunjaya P, Seider MI, Stinnett S, Scheffler A; Ocular Oncology Study Consortium. Association between tumor regression rate and gene expression profile after iodine 125 plaque radiotherapy for uveal melanoma. *Ophthalmology*. 2017;124(10):1532–9. <https://doi.org/10.1016/j.ophtha.2017.04.013>
- 22 Chiam PJ, Coupland SE, Kalirai H, Groenewald C, Heimann H, Damato BE. Does choroidal melanoma regression correlate with chromosome 3 loss after ruthenium brachytherapy? *Br J Ophthalmol*. 2014;98(7):967–71. <https://doi.org/10.1136/bjophthalmol-2013-304472>

- 23 Fang R, Wang H, Li Y, Liu YM, Wei WB. Regression patterns of uveal melanoma after iodine-125 plaque brachytherapy. *BMC Ophthalmol*. 2021;21(1):137. <https://doi.org/10.1186/s12886-021-01898-3>
- 24 Salvi SM, Aziz HA, Dar S, Singh N, Hayden-Loreck B, Singh AD. Uveal melanoma regression after brachytherapy: relationship with chromosome 3 monosomy status. *Ocul Oncol Pathol*. 2017;3(2):87–94. <https://doi.org/10.1159/000451001>
- 25 Miguel D, Saornil MA, de Frutos JM, García-Álvarez C, Alonso P, Diezhandino P. Regression of posterior uveal melanoma following iodine-125 plaque radiotherapy based on pre-treatment tumor apical height. *J Contemp Brachytherapy*. 2021;13(2):117–25. <https://doi.org/10.5114/jcb.2021.105278>
- 26 Field MG, Durante MA, Decatur CL, Tarlan B, Oelschlager KM, Stone JF, et al. Epigenetic reprogramming and aberrant expression of PRAME are associated with increased metastatic risk in Class 1 and Class 2 uveal melanomas. *Oncotarget*. 2016;7(37):59209–19. <https://doi.org/10.18632/oncotarget.10962>
- 27 Field MG, Decatur CL, Kurtenbach S, Gezgin G, van der Velden PA, Jager MJ, et al. PRAME as an independent biomarker for metastasis in uveal melanoma. *Clin Cancer Res*. 2016;22(5):1234–42. <https://doi.org/10.1158/1078-0432.CCR-15-2071>
- 28 Lane AM, Oxenreiter MM, Hashmi M, Aronow ME, Trofimov AV, Shih HA, et al. A comparison of treatment outcomes after standard dose (70 Gy) versus reduced dose (50 Gy) proton radiation in patients with uveal melanoma. *Ophthalmol Retina*. 2022;6(11):1089–97. <https://doi.org/10.1016/j.oret.2022.05.006>
- 29 Kujala E, Mäkitie T, Kivelä T. Very long-term prognosis of patients with malignant uveal melanoma. *Invest Ophthalmol Vis Sci*. 2003;44(11):4651–9. <https://doi.org/10.1167/iiov.03-0538>
- 30 Onken MD, Worley LA, Char DH, Augsburger JJ, Correa ZM, Nudleman E, et al. Collaborative Ocular Oncology Group report number 1: prospective validation of a multi-gene prognostic assay in uveal melanoma. *Ophthalmology*. 2012;119(8):1596–603. <https://doi.org/10.1016/j.opht.2012.02.017>
- 31 Lattman J, Kroll S, Char DH, Ghazvini S, Frigillana H, O'Brien JM, et al. Cell cycling and prognosis in uveal melanoma. *Clin Cancer Res*. 1995;1(1):41–7.
- 32 Augsburger JJ, Gonder JR, Amsel J, Shields JA, Donoso LA. Growth rates and doubling times of posterior uveal melanomas. *Ophthalmology*. 1984;91(12):1709–15. [https://doi.org/10.1016/s0161-6420\(84\)34088-x](https://doi.org/10.1016/s0161-6420(84)34088-x)
- 33 Pacheco R, Stock H. Effects of radiation on bone. *Curr Osteoporos Rep*. 2013;11(4):299–304. <https://doi.org/10.1007/s11914-013-0174-z>
- 34 Borrego-Soto G, Ortiz-López R, Rojas-Martínez A. Ionizing radiation-induced DNA injury and damage detection in patients with breast cancer. *Genet Mol Biol*. 2015;38(4):420–32. <https://doi.org/10.1590/S1415-475738420150019>
- 35 Harbour JW, Correa ZM, Scheffler AC, Mruthyunjaya P, Materin MA, Aaberg TA Jr, et al. 15-Gene expression profile and PRAME as integrated prognostic test for uveal melanoma: first report of collaborative ocular Oncology group study No. 2 (COOG2.1). *J Clin Oncol*. 2024;42(28):3319–29. <https://doi.org/10.1200/JCO.24.00447>
- 36 Rashid M, Heikkonen J, Kivelä T. Tumor regression after brachytherapy for choroidal melanoma: reduction of thickness and cross-sectional area by shape and regression pattern. *Invest Ophthalmol Vis Sci*. 2015;56(4):2612–23. <https://doi.org/10.1167/iiov.14-16322>